MILITARY SPECIFICATION
FIRING MECHANISM, CARTRIDGE ACTUATED, 29378-1
REMANUFACTURED METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Firing Mechanisms P/N 29378-1. The assembly will be ultimately loaded and used to actuate the canopy counter-balance jettison and cinch-down cylinder assembly in the TF102 Aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION
Military

MIL-A-?550	- Ammunition and Special Weapons, General Specification for

STANDARDS
Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings

DRAWINGS
Munitions Command

C9224217	- Container, Ammunition, Fiber for Firing Mechanism Assembly, P/N 29378-1
C9224720	- Box, Packing, Ammunition, for Firing Mechanism Assembly, P/N 29378-1
IEL29378-1	- Inspection Equipment List for Firing Mechanism, Cartridge Actuated, 29378-1 Assembly

Talley Industries (Code Ident. No. 12116)

29378-1	- Firing Mechanism, Cartridge Actuated, Assembly

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(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply:

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016).

3. REQUIREMENTS

3.1 General - Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Material - The contractor shall be responsible for selecting from the classification list contained in the raw material intended for use in manufacture.

3.3 Mechanical properties - Mechanical properties required by the drawing shall be uniform throughout the p , within specified tolerances. Requirements for all steel parts shall be achieved by quenching and tempering.

3.4 Metal parts assembly - Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing and the following:

3.4.1 The remanufactured Firing Mechanism metal parts assembly shall contain new parts in accordance with Table I:

TABLE I

Pin, Shear	[illegible]
O-Ring	AN 6227-7
Set Screw	AN 565D8H2
Pellet (2)	[illegible]

3.5 Protective finish - Protective finish shall be in accordance with the applicable specification and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.6 Residual magnetism - The metal parts assembly shall not deflect the indicator of the compass more than three degrees in either direction when passed in front of the compass at a distance of six inches.

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3.7 Workmanship - The requirements for workmanship are as specified on the applicable, referenced specifications and the following:

3.7.1 Metal defects - All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.7.2 Burr - No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter - No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions - Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection - Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions

4.2.1 Lot

4.2.1.1 Submission of product - The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot - For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 firing mechanisms maximum, exclusive of test samples, provided the homogeneity of remanufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.2.2 Examination - The roughness comparison specimens prescribed by ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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4.2.2.1 Reclaimed parts - Examination of reusable parts, which require no reprocessing shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be rejected..

4.2.2.2 Reprocessed parts - One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE III

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects Number of Defects in Class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14

Sample Sizes for Major Defects Number of Defects in Class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

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4.2.2.3 Head, Drawing 29308-1 covering a detail of Drawing 29378-1

Detail		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of large thread	Gage
102.	Major diameter of large thread	Gage
103.	Pitch diameter of small thread	Gage
104.	Major diameter of small thread	Gage
105.	Pitch diameter of set screw thread	Gage
106.	Major diameter of set screw thread	Gage
107.	Shear pin hole	Gage
108.	Metal defective	Visual
109.	Protective finish inadequate or defective	Visual
110.	Surface finish improper	Visual
111.	Burr	Visual
112.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.4 Chamber, Drawing 29309-0 covering a detail of Drawing 29378-1

CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of large thread	Gage
102.	Minor diameter of large thread	Gage
103.	Pitch diameter of small thread	Gage
104.	Minor diameter of small thread	Gage
105.	Thread damaged	Visual
106.	Metal defective	Visual
107.	Protective finish inadequate or defective	Visual
108.	Surface finish improper	Visual
109.	Burr	Visual
110.	Foreign matter	Visual
MINOR		
	None defined	

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4.2.2.5 Container, Ammunition, Fiber, Drawing C9224217

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Gap between body and cover, max.	Manual
202.	Contents loose	Manual
203.	Metal ends loose or distorted	Manual
204.	Cut or damaged through all impregnated layers	Visual
205.	Marking, missing, incorrect or illegible	Visual
206.	Tape incomplete or badly wrinkled	Visual
207.	Length of tear tab, min.	Visual
208.	Color of tape incorrect	Visual

4.2.2.6 Box, Packing, Ammunition, Drawing C9224720

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual
202.	Hardware missing, loose, broken or improperly engaged	Manual and Visual
203.	Strapping missing, loose, broken or improperly assembled	Manual and Visual

4.2.3 Inspection - The following tests shall be performed in accordance with 4.3.

4.2.3.1 Protective finish, replated components - The number of specimens and acceptance criteria for protective finish shall be in accordance with the applicable military standard for the type of protective finish utilized in the rebuild process.

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4.2.3.2 Residual magnetism - Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.6 shall be cause for rejection of the assembly.

4.2.4 Inspection equipment - The examination and tests shall be made using equipment approved by the Government.

4.3 Test methods and procedures

4.3.1 Protective finish, replated components - Test procedures for protective finish shall be in accordance with the applicable military standards for the type of protective finish utilized in the rebuild process.

4.3.2 Residual magnetism - The assembly shall be tested for excessive residual magnetism, in an area free of local magnetic effects, by placing the assembly six inches from and at the same height as the compass in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly past the compass. This procedure shall be repeated for each 90 degree rotation of the firing mechanism.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A - Packing and marking shall be in accordance with drawings C9224217 and C9224720.

6. NOTES

6.1 Ordering data - Procurement documents should specify the title, number, and date of this specification.

6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 Remanufactured exterior components are listed in Table IV:

TABLE IV

Name of Part	Part Number
Head	29308-1
Chamber	29309-0

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6.4 When remanufacturing and loading of assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

Custodian:

Army - MU

Preparing activity:

Army - MU - (FA)

Project No. 1377-A413

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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